



魅力统计

Fascinating Statistics

第六章 常用监督学习方法I

Chapter 6 Common Supervised Learning Methods I

Introduction

- In the following two chapters, we will learn basic statistical models / machine learning models.
- There is no clear distinction between statistical learning and machine learning.



2012 年《统计学习方法 (第 1 版)》出版, 内容涵盖监督学习的主要方法, 2019 年第 2 版出版, 增加了无监督学习的主要方法, 都属于传统机器学习。在这段时间里, 机器学习领域发生了巨大变化, 深度学习在人工智能各个应用方向取得了巨大突破, 成为机器学习的主流技术, 彻底改变了机器学习的面貌。有些读者希望能看到与之前风格相同的讲解深度学习的书籍, 这也触发了作者在原来《统计学习方法》的基础上增加深度学习内容的想法 (计划今后再增加强化学习)。从 2018 年开始, 历时 3 年左右, 完成了深度学习的写作。

考虑到内容的变化, 现将书名更改为《机器学习方法》。第 1 篇监督学习和第 2 篇无监督学习基本为原来的内容, 增加第 3 篇深度学习, 希望对读者有所裨益。传统机器学习是深度学习的基础, 所以将这些内容放在一本书里讲述也有其合理之处。虽然深度学习目前是大家关注的重点, 但传统机器学习仍然有其不容忽视的地位。事实上, 传统机器学习和深度学习各自有更适合的应用场景, 比如, 深度学习长于大数据、复杂问题的预测, 特别是人工智能的应用; 传统机器学习善于小数据、相对简单问题的预测。

本书的定位是讲解机器学习的基本内容, 并不完全是入门书。介绍的内容都是最基本的, 在这种意义上适合初学者。但主旨是把最重要的原理和方法做系统的总结, 方便大家经常阅读和复习。在写第 3 篇的时候也接受大家对第 1 篇和第 2 篇的反馈意见, 在力求文字简练清晰的同时, 也确保叙述的详明了, 以方便读者理解。在各章方法的导入部分适当增加了背景和动机的介绍。

第 3 篇中使用的数学符号与第 1 篇和第 2 篇有一定的对应关系, 但由于深度学习的特点也有一些改变, 也都能自成体系。将符号完全统一于一个框架内还需要做大量的工作, 希望在增加第 4 篇强化学习之后再进行处理。

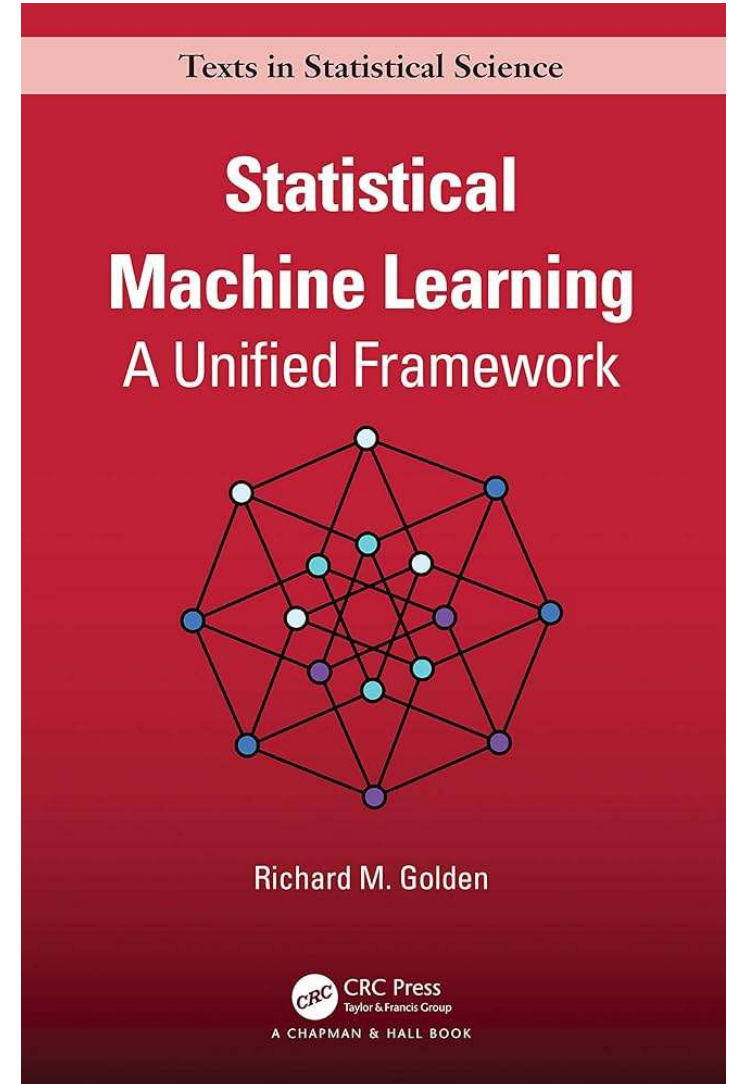
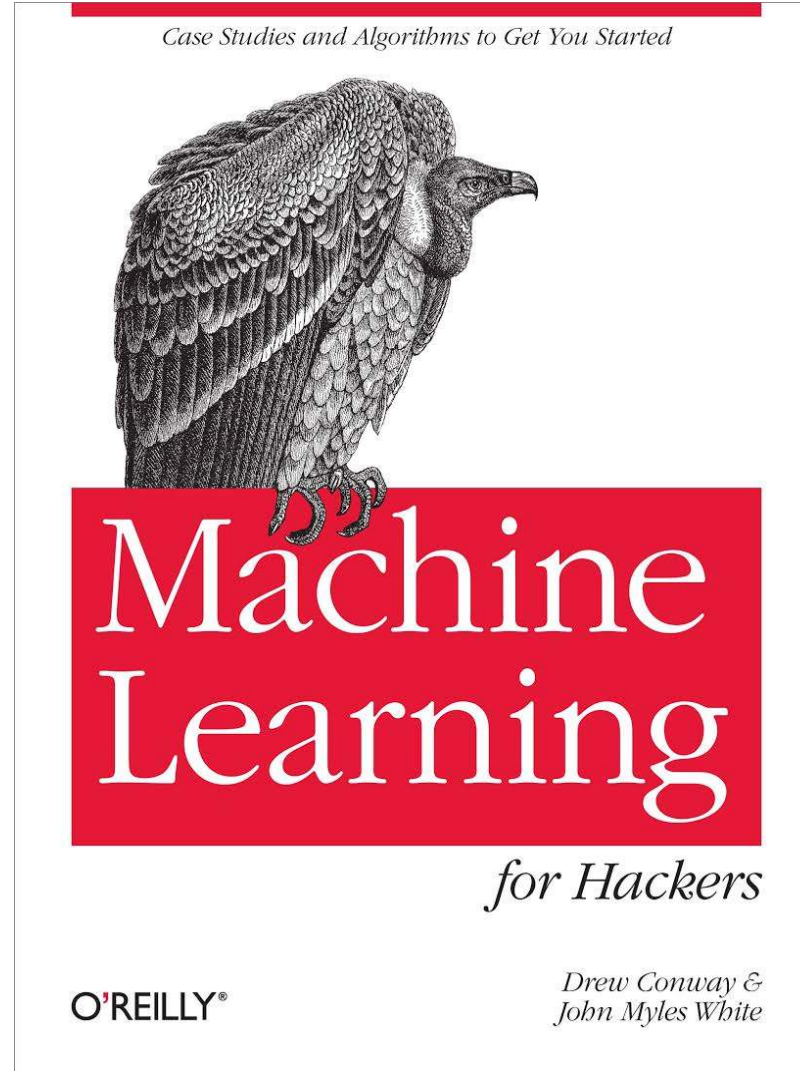
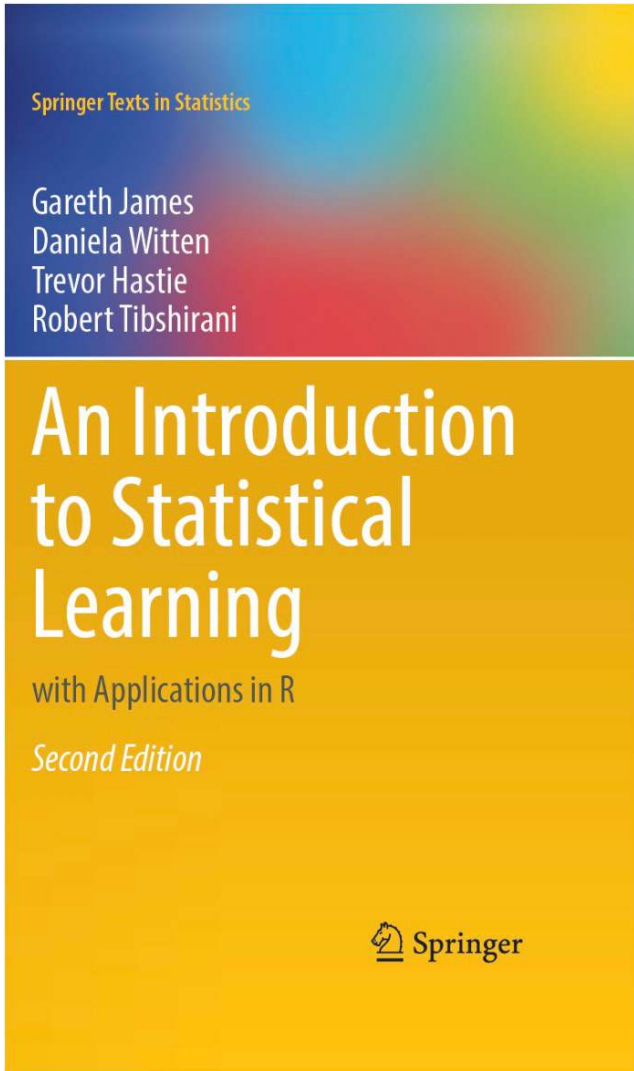
对第 3 篇的原稿, 郑诗源、张新松等帮助做了校阅, 对一些章节的内容提出了宝贵的意见。责任编辑王倩也为本书的出版做了大量工作。在此对他们表示衷心的感谢。

李 航

2021 年 5 月 27 日



Introduction



Introduction

- GPT-4's answer on the difference between statistical models and machine learning models:

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What's the difference between statistical models and machine learning models?



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Statistical models and machine learning models are both tools for analyzing data and making predictions, but they differ in their approaches, methodologies, and applications. Here's a detailed comparison:

Definitions

- **Statistical Models** are a part of statistical methods that embody a set of **statistical assumptions** concerning the generation of some sample data and similar data from a larger population. These models are traditionally used for hypothesis testing, estimation, and making inferences about relationships within the data based on **probabilistic theories**.
- **Machine Learning Models** are based on the concept of learning from data, improving their accuracy over time as they are exposed to more data. Unlike statistical models that are primarily developed from a theoretical understanding of the data generating process, machine learning models focus on **making predictions or decisions** without being explicitly programmed to perform the task.



Introduction

Example Models

- **Statistical Models:** Linear regression, logistic regression, Cox proportional hazards model, ANOVA, etc.
- **Machine Learning Models:** Neural networks, decision trees, support vector machines, random forests, gradient boosting machines, etc.

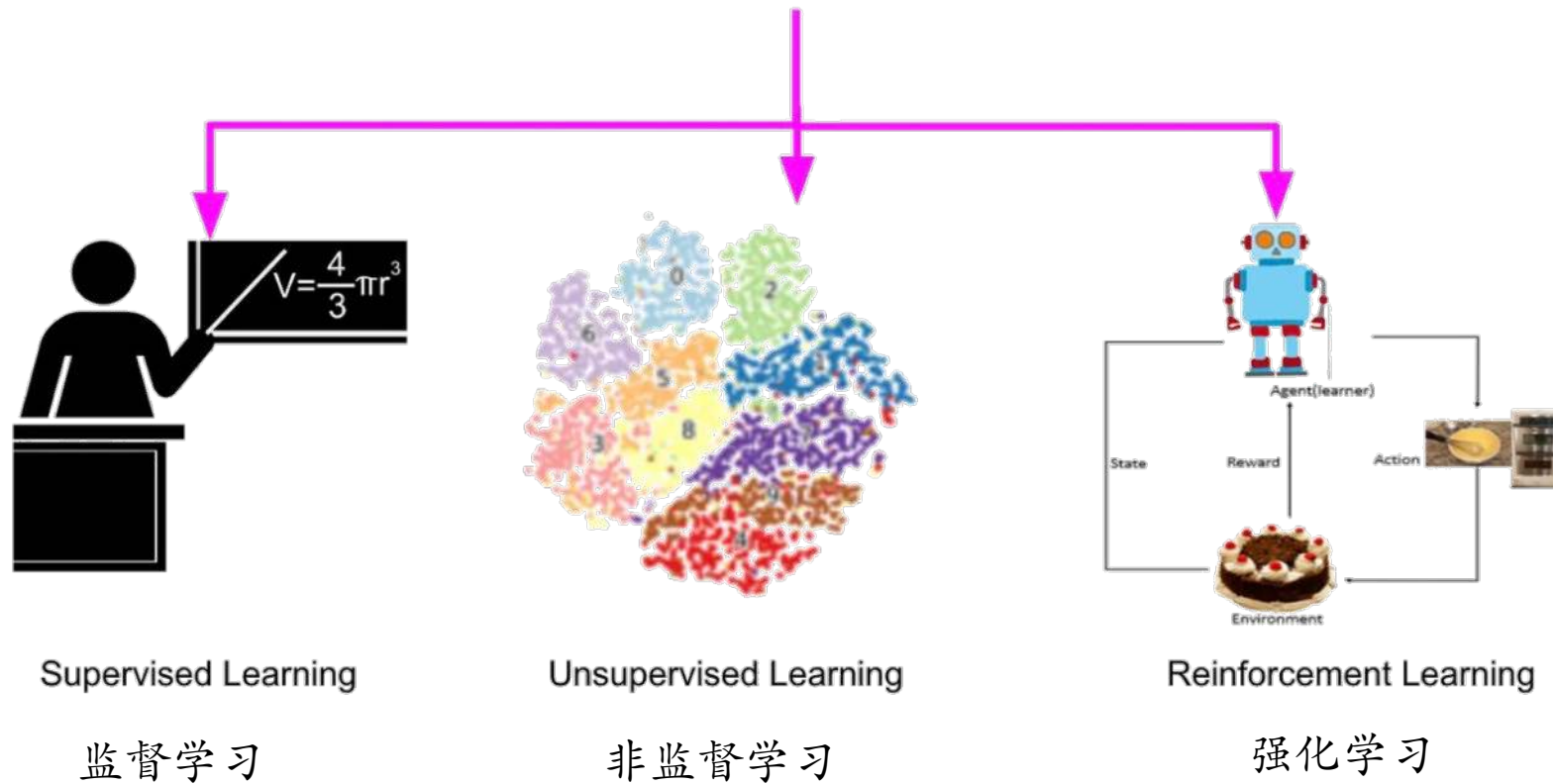
Conclusion

In summary, statistical models emphasize understanding and interpreting the relationships in the data based on a theoretical framework, while machine learning models focus on maximizing predictive accuracy, often at the expense of interpretability. The choice between using a statistical model or a machine learning model depends on the goals of the analysis, the nature of the data, and the importance of interpretability versus prediction accuracy in the specific context.



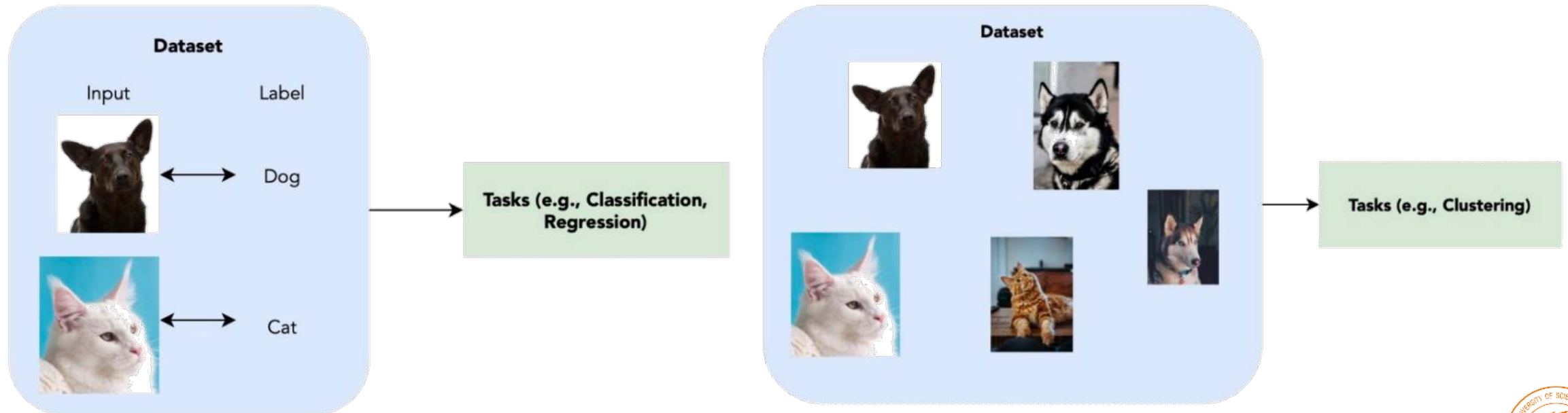
Introduction

- Statistical learning / machine learning can generally be categorized into the following three types:



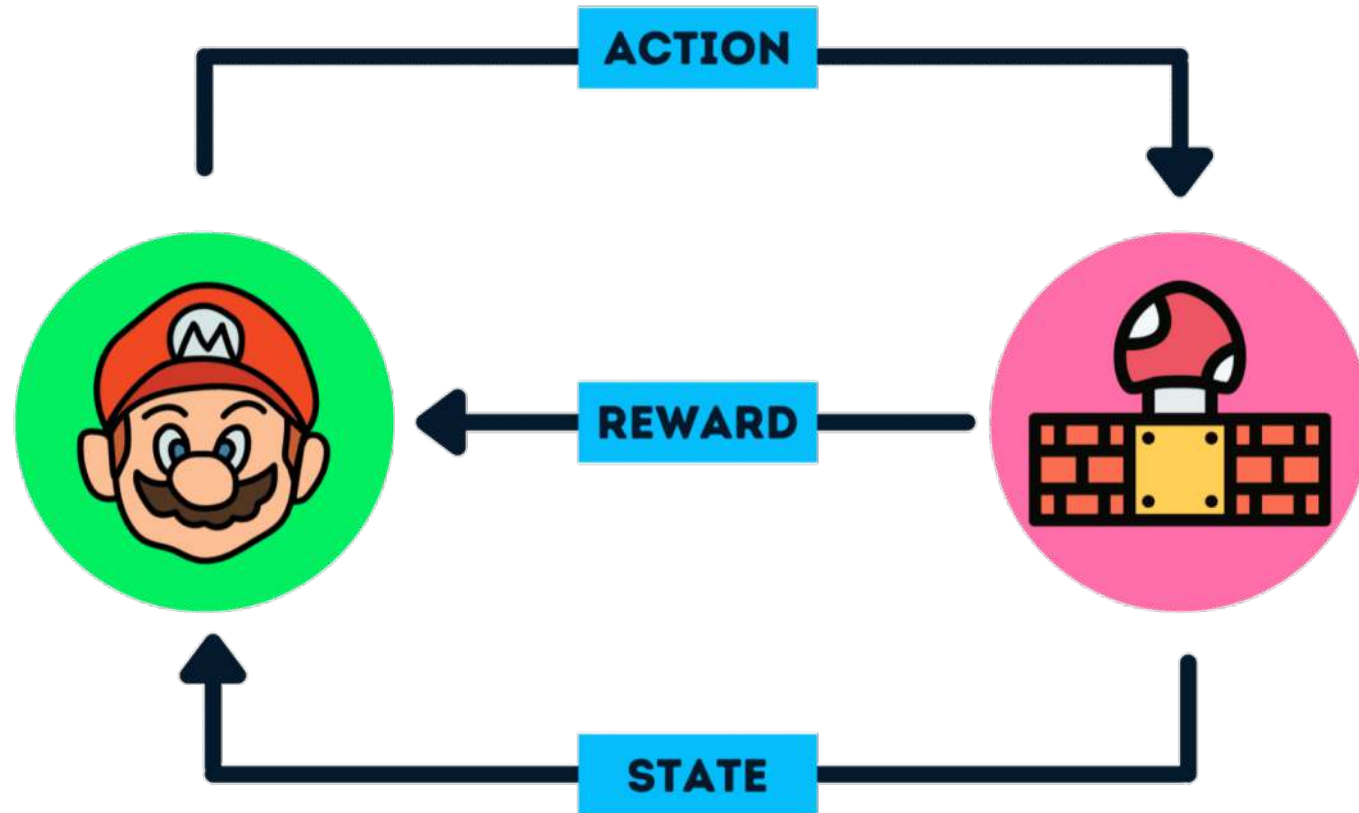
Introduction

- Supervised learning (监督学习) refers to learning the correspondence between input and output from labeled data (标注数据).
- Unsupervised learning (非监督学习) refers to learning statistical patterns or potential structures from unlabeled data (无标注数据).



Introduction

- Reinforcement learning (强化学习) refers to the process of an agent learning the optimal behavior strategy through interaction with the environment.



Chapter 6 Common Supervised Learning Methods I

- 6.1 Linear Regression Model
- 6.2 Logistic Regression Model
- 6.3 k-Nearest Neighbor Algorithm
- 6.4 Naïve Bayes Classifier
- 6.5 Support Vector Machine
- 6.6 General Pattern of Supervised Learning



6.1 Linear Regression Model

- In 1990, economist **William Sharpe** (威廉·夏普) won the Nobel Prize in Economics for proposing the **Capital Asset Pricing Model** (CAPM, 资本资产定价模型).
- The model provides the relationship between the expected return of an asset i and the expected return of the market portfolio:

$$E(R_i) = R_f + \beta_i[E(R_m) - R_f].$$

- In 2013, **Eugene Fama** (尤金·法马) won the Nobel Prize in Economics for proposing the **Fama-French Three Factor Model** (Fama-French 三因子模型).
- It is a revision of the CAPM:

$$E(R_i) = R_f + \beta_{1i}[E(R_m) - R_f] + \beta_{2i}E(SMB) + \beta_{3i}E(HML),$$

- **SMB** (small minus big) is the **market capitalization factor** (市值因子), i.e., the historic excess returns of small-cap companies relative to large-cap companies;
- **HML** (high minus low) is the **book-to-market ratio factor** (账面市值比因子), i.e., the historic excess returns of value stocks over growth stocks.

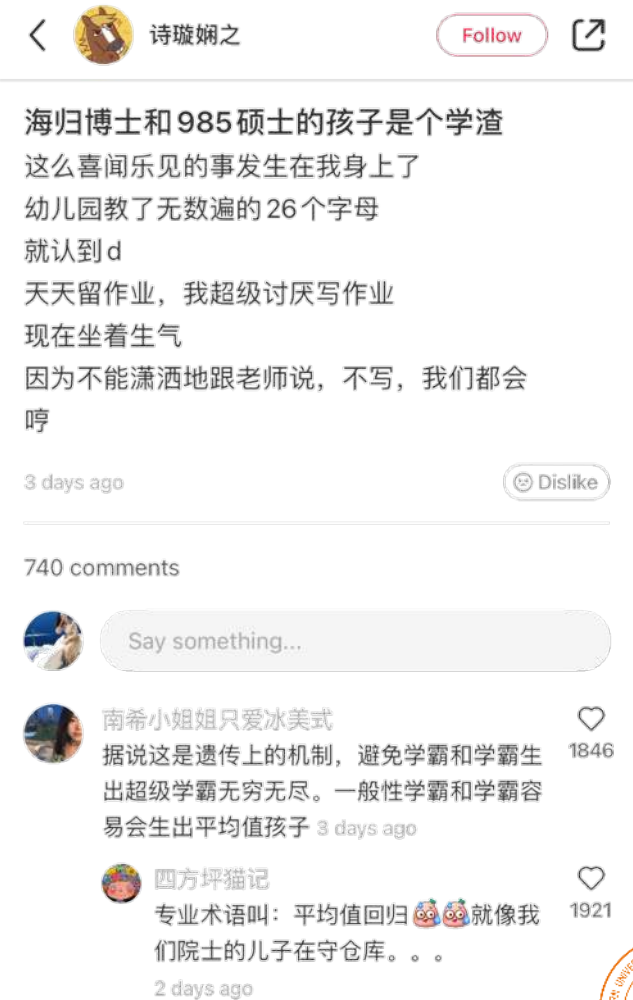


6.1 Linear Regression Model

- Both CAMP and Fama-French Three Factor Model are **regression models (回归模型)** with linear structures.
- Simply put, regression model is used to analyze the relationship between the expectation of a dependent variable (**因变量/响应变量, response variable**) and some independent variables (**自变量/解释性变量, explanatory variables**):

$$E(Y|X = x) = f(x).$$

- The term “regression” was first proposed by **Francis Galton** when he found that children of tall parents tended to be tall, but **on average** not as tall as their parents.
- He describe this phenomenon as “**regression to the mediocrity (向均值回归)**”. There seems to be an invisible hand in nature that controls human height within a relatively stable range.
- Similar phenomena are everywhere in daily life.

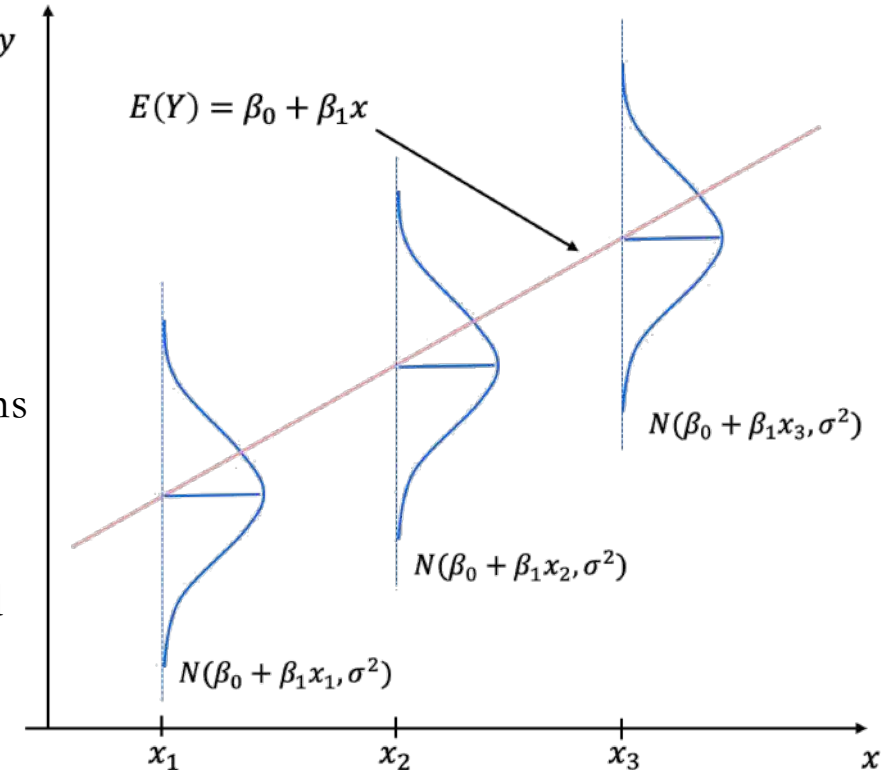


6.1 Linear Regression Model

- linear regression model (线性回归模型, or linear model, 线性模型) is a type of regression model.
- Take the simple linear regression model (简单线性回归模型) as an example:

$$Y = E(Y|X) + \varepsilon = \beta_0 + \beta_1 X + \varepsilon.$$

- β_0 (intercept, 截距) and β_1 (slope, 斜率) are parameters to be estimated, called regression coefficients (回归系数).
- ε is the random error (随机误差) which refers to the fluctuations in Y that cannot be explained by X and is typically assumed to follow a normal distribution, i.e., $\varepsilon \sim N(0, \sigma^2)$.
- The goal is to estimate parameters β_0 and β_1 based on observed data $\{(x_i, y_i), i = 1, 2, \dots, n\}$.



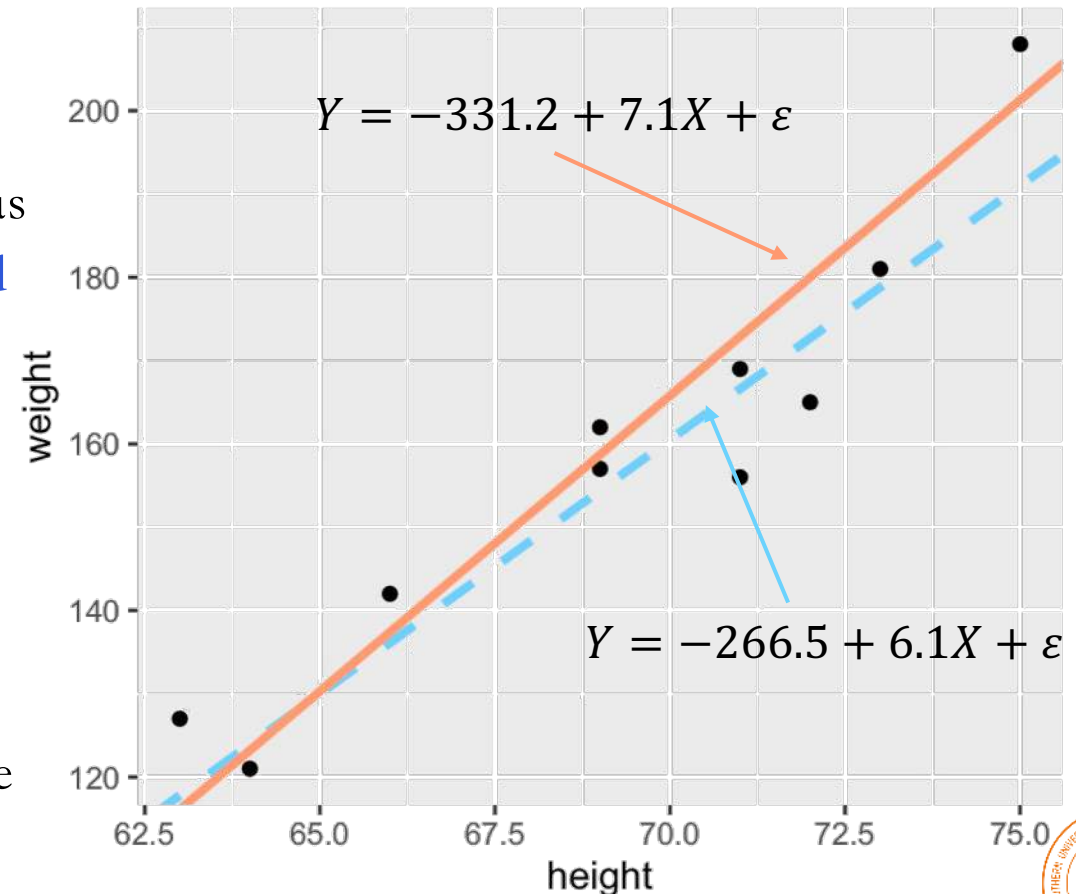
Example 6.1

How to estimate the intercept and slope in a simple linear regression model of weight vs. height based on the height and weight of 10 people?



6.1 Linear Regression Model

- Draw the scatterplot of weight vs. height, then a linear model is a straight line on the plot.
- Which one is better, the blue dotted line or the orange solid line?
- Denote the intercept and slope of the fitted line as $\hat{\beta}_0$ and $\hat{\beta}_1$, then $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ is called the **fitted value** (拟合值) or **predicted value** (预测值) of Y_i .
- A natural idea is to consider the **residual** (残差, also called the **prediction error**, 预测误差):
$$e_i = y_i - \hat{y}_i.$$
- A well-fitted line should have small residual.
- Taking into account different individuals, the line with smaller $\sum (y_i - \hat{y}_i)^2$ is better.



6.1 Linear Regression Model

$$Y = -331.2 + 7.1X + \varepsilon$$

i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	116.1	10.9	118.81
2	64	121	123.2	-2.2	4.84
3	66	142	137.4	4.6	21.16
4	69	157	158.7	-1.7	2.89
5	69	162	158.7	3.3	10.89
6	71	156	172.9	-16.9	285.61
7	71	169	172.9	-3.9	15.21
8	72	165	180.0	-15.0	225.00
9	73	181	187.1	-6.1	37.21
10	75	208	201.3	6.7	44.89
					766.5

$$Y = -266.5 + 6.1X + \varepsilon$$

i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	120.139	6.8612	47.076
2	64	121	126.276	-5.2764	27.840
3	66	142	138.552	3.4484	11.891
4	69	157	156.964	0.0356	0.001
5	69	162	156.964	5.0356	25.357
6	71	156	169.240	-13.2396	175.287
7	71	169	169.240	-0.2396	0.057
8	72	165	175.377	-10.3772	107.686
9	73	181	181.515	-0.5148	0.265
10	75	208	193.790	14.2100	201.924
					597.4

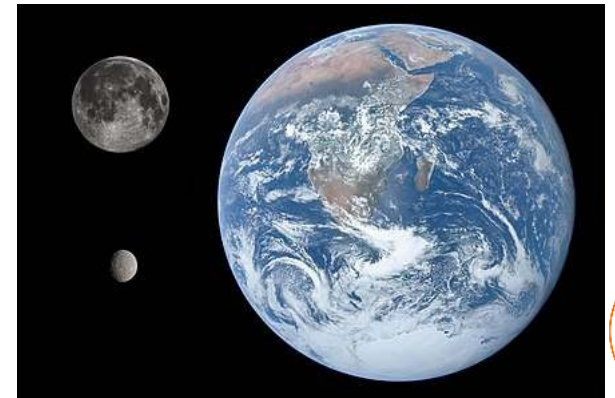
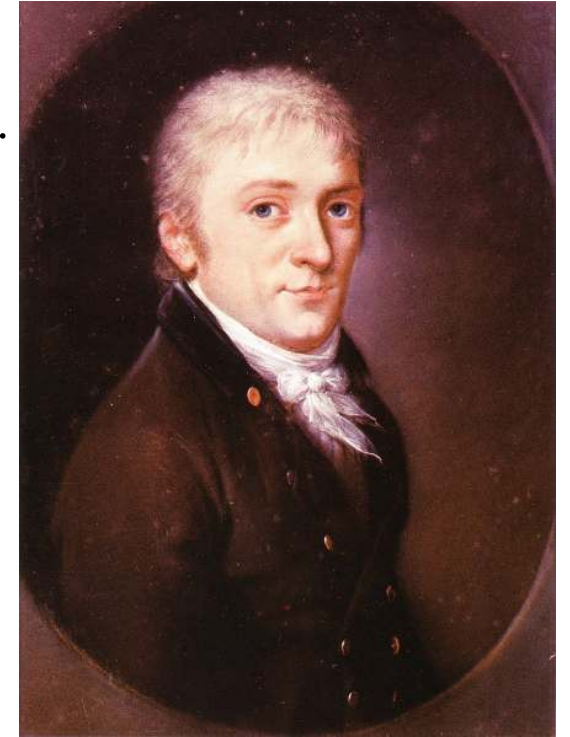


6.1 Linear Regression Model

- Naturally, the best fitting line is the one with the smallest $\sum(y_i - \hat{y}_i)^2$.
- Idea of the least squares method (LSE, 最小二乘法): minimize the sum of squared error (SSE, 误差平方和, also called residual sum of squares, RSS, 残差平方和):

$$\text{SSE}(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \Rightarrow \hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \text{SSE}(\boldsymbol{\beta}).$$

- $\hat{\beta}_0$ and $\hat{\beta}_1$ obtained in this way is termed the least squares estimator / estimate (LSE, 最小二乘估计量/值).
- As early as 1801, Gauss used the least squares method to identify the orbit of Ceres.
- Why minimize $\sum(y_i - \hat{y}_i)^2$ instead of $\sum|y_i - \hat{y}_i|$?
- Under the linear model, the LSE of β_0 and β_1 has explicit expression.



6.1 Linear Regression Model

$$\begin{cases} \frac{\partial \text{SSE}}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \frac{\partial \text{SSE}}{\partial \beta_1} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i = 0 \end{cases} \Rightarrow \begin{cases} n\hat{\beta}_0 + n\bar{x}\hat{\beta}_1 = n\bar{y} \\ n\bar{x}\hat{\beta}_0 + \sum x_i^2 \hat{\beta}_1 = \sum x_i y_i \end{cases}$$
$$\Rightarrow \hat{\beta}_1 = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Example 6.1 (Continued)

Please calculate the LSE of the intercept and slope in a linear regression model of weight vs. height based on the data of the following 10 people:

i	1	2	3	4	5	6	7	8	9	10
height x_i	63	64	66	69	69	71	71	72	73	75
weight y_i	127	121	142	157	162	156	169	165	181	208



6.1 Linear Regression Model

Solution

First, calculate the average height and weight: $\bar{x} = 69.3, \bar{y} = 158.8$. Then obtain

i	1	2	3	4	5	6	7	8	9	10
$x_i - \bar{x}$	-6.3	-5.3	-3.3	-0.3	-0.3	1.7	1.7	2.7	3.7	5.7
$y_i - \bar{y}$	-31.8	-37.8	-26.8	-1.8	3.2	-2.8	10.2	6.2	22.2	49.2
$(x_i - \bar{x})(y_i - \bar{y})$	200.34	200.34	55.44	0.54	-0.96	-4.76	17.34	16.74	82.14	280.44
$(x_i - \bar{x})^2$	39.69	28.09	10.89	0.09	0.09	2.89	2.89	7.29	13.69	32.49

Consequently:

$$\sum(x_i - \bar{x})(y_i - \bar{y}) = 847.6, \quad \sum(x_i - \bar{x})^2 = 138.1.$$

$$\Rightarrow \hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{847.6}{138.1} \approx 6.1376, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 158.8 - \frac{847.6}{138.1} \times 69.3 = -266.5344.$$

Therefore, the fitted model is $\hat{y} = -266.5344 + 6.1376x$.



6.1 Linear Regression Model

Solution

Applying the `lm()` function in R, it is easy to obtain the estimated regression coefficients:

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -266.5344    51.0320  -5.223   8e-04 ***
height       6.1376     0.7353   8.347 3.21e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Note: considering that Y_i 's are random variables, then $\hat{\beta}_0$ and $\hat{\beta}_1$ are also random variables. It can be shown that they are unbiased estimators of β_0 and β_1 under the model assumptions.
- Other than the regression coefficients, we are also interested in estimating σ^2 .
 - σ^2 is the variance of the random errors, we naturally think of estimating it based on the residuals $e_i = y_i - \hat{y}_i$. An unbiased estimator of σ^2 is:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$



6.1 Linear Regression Model

Solution

To estimate σ^2 , we calculate $\hat{y}_i$ based on the fitted model $\hat{y}_i = -266.5344 + 6.1376x_i$:

i	1	2	3	4	5	6	7	8	9	10
x_i	63	64	66	69	69	71	71	72	73	75
y_i	127	121	142	157	162	156	169	165	181	208
$\hat{y}_i$	120.12	126.27	138.55	156.96	156.96	169.24	169.24	175.37	181.51	193.79
$(y_i - \hat{y}_i)^2$	47.14	27.79	11.92	0.0016	25.40	175.17	0.055	107.59	0.261	202.05

So that we obtain $\sum_{i=1}^n (y_i - \hat{y}_i)^2 = 597.386$, and consequently:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{597.386}{8} = 74.6733 \Rightarrow \hat{\sigma} = \sqrt{74.6733} \approx 8.6414.$$

The estimated value of $\hat{\sigma}$ is reported in the output of the `lm()` function in R:

```
Residual standard error: 8.641 on 8 degrees of freedom
Multiple R-squared: 0.897, Adjusted R-squared: 0.8841
F-statistic: 69.67 on 1 and 8 DF, p-value: 3.214e-05
```



Chapter 6 Common Supervised Learning Methods I

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6.2 Logistic Regression Model

- As a regression model, the logistic regression model is **mainly used for binary classification**.
 - Suppose that the response variable Y takes only two possible values, i.e., $Y = 0$ and $Y = 1$. We want to study the relationship between Y and some explanatory variables $\mathbf{X} = (X_1, X_2, \dots, X_p)$.
 - First, we think of the regression model $E(Y|\mathbf{X}) = f(\mathbf{X})$.
 - As the linear regression model, we would like to try a linear form for $f(\mathbf{X})$, i.e., $f(\mathbf{X}) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$.
 - But there is an obvious problem with this model: the value of $E(Y|\mathbf{X}) = P(Y = 1|\mathbf{X}) \stackrel{\text{def}}{=} p$ is limited to between 0 and 1, while the linear combination of explanatory variables may not meet this restriction.
 - A natural way to solve this problem is to: transform p into a value with no restriction ($-\infty$ to ∞), or equivalently map $\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$ to the interval $[0, 1]$.
 - What transformations can you think of?

Logit

$$g(p) = \log\left(\frac{p}{1-p}\right), \quad g(p) = \Phi^{-1}(p), \quad g(p) = \log[-\log(1-p)].$$

Probit

Complementary log-log



6.2 Logistic Regression Model

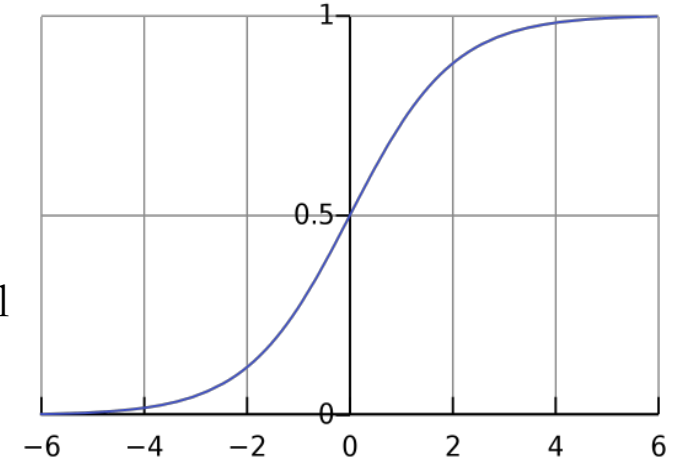
- The logit transformation is the most commonly used one, its inverse function $g^{-1}(x) = e^x / (1 + e^x) = 1 / (1 + e^{-x})$ is called the **logistic function**. Therefore, the model based on this transformation is termed the **logistic regression** (逻辑斯蒂回归/逻辑回归):

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \Rightarrow p = \frac{\exp(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)}{1 + \exp(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)}.$$

- The logistic function is a common S-shaped function with a wide range of applications, e.g., ecology, demography, economics, sociology, etc.
 - In the early 19th century, people were interested in studying population growth, Adolphe Quetelet (阿道夫·凯特勒) represent the total population at time t by $W(t)$, then the growth rate is $dW(t)/dt$.
 - He found that the population growth rate is linearly related to the total population:

$$\frac{dW(t)}{dt} = \beta W(t).$$

- Solving this differential equation yields $\log W(t) = \beta_0 + \beta t \Rightarrow W(t) = \exp(\beta_0 + \beta t)$.



6.2 Logistic Regression Model

- The logistic function:

- The equation suggests that the total population grows exponentially over time. However, the resources on earth are limited so that it is impossible for the population to grow endlessly.
- Adolphe found this question very interesting and ask his student Pierre to study it further.
- Pierre believed that a restriction function $h(W(t))$ can be added to control the population growth:

$$\frac{dW(t)}{dt} = \beta W(t) - h(W(t)).$$

- After countless attempts and failures, Pierre discovered that the ideal restriction function is a quadratic function, leading to

$$\frac{dW(t)}{dt} = \tilde{\beta} W(t)[N - W(t)],$$

- where N represents the maximum total population that the environment can sustain, $\tilde{\beta} = \beta/N$. Solving the differential equation, Pierre obtain

$$W(t) = \frac{N \exp(\beta_0 + \beta t)}{1 + \exp(\beta_0 + \beta t)}.$$

- Pierre named this function the logistic function in a paper published in 1845.



6.2 Logistic Regression Model

- Under the logistic regression model, how to estimate the parameter $\beta_0, \beta_1, \dots, \beta_p$ based on the observed data $\{(y_i, x_{i1}, x_{i2}, \dots, x_{ip}), i = 1, 2, \dots, n\}$?

Example 6.2

- The sinking of Titanic is one of the most infamous shipwrecks in history.
- On April 15th, 1912, the Titanic sank after colliding with an iceberg, killing 1,500+ out of 2,224 passengers and crew.
- The sensational tragedy shocked the international community and led to better safety regulations for ships.
- One of the reasons that the shipwreck led to such loss of life was that there were not enough lifeboats.



6.2 Logistic Regression Model

Example 6.2

- From the film, it seems that some groups of people were more likely to survive than others, such as women, children, and the upper-class.
- Survival prediction on the Titanic is one of the most classical and beginner-friendly data analysis projects.
- We will use the dataset from the Kaggle competition platform to analyze “what kind of people were more likely to survive”.
- First six records of the Titanic dataset (after some preprocessing) are given below:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Fare	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	7.25	S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	71.2833	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	7.925	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	53.1	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	8.05	S
6	0	3	Moran, Mr. James	male	39	0	0	8.4583	Q

Data source: <https://www.kaggle.com/competitions/titanic>



6.2 Logistic Regression Model

- Regression coefficients in the logistic regression model can be estimated using the maximum likelihood method.
- The likelihood function based on the observed data $\{(y_i, x_{i1}, x_{i2}, \dots, x_{ip}), i = 1, 2, \dots, n\}$ ($y_i \in \{0, 1\}$) is

$$\begin{aligned} L(\beta_0, \beta_1, \dots, \beta_p) &= \prod_{i=1}^n p^{y_i} (1 - p)^{1-y_i} \\ &= \prod_{i=1}^n \left\{ \frac{\exp(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}{1 + \exp(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)} \right\}^{y_i} \left\{ \frac{1}{1 + \exp(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)} \right\}^{1-y_i}. \end{aligned}$$

- Then the log-likelihood function is

$$\ell(\beta_0, \beta_1, \dots, \beta_p) = \sum_{i=1}^n \{y_i(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p) - \ln[1 + \exp(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)]\}.$$

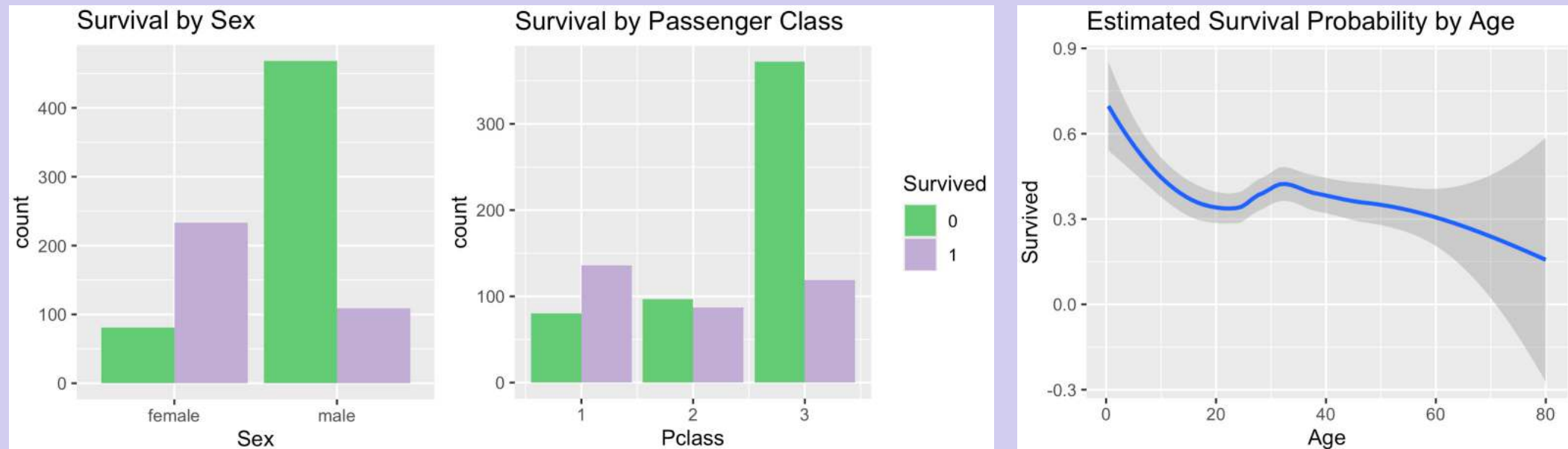
- The difficulty is that, unlike the linear regression model, $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ cannot be obtained explicitly by taking the derivative of $\ell(\beta_0, \beta_1, \dots, \beta_p)$.
- Optimization algorithm, such as [gradient descent](#) (梯度下降法) and [Newton's method](#) (牛顿法), need to be applied.



6.2 Logistic Regression Model

Example 6.2 - Solution

- First, see whether survival is related to sex, passenger class, and age through data visualizations.



- Then let $p = P(Y = 1|\mathbf{X})$ be the probability of survival, try to fit the following logistic regression model:

$$\text{logit}(p) = \beta_0 + \beta_1 I(\text{Sex} = \text{male}) + \beta_2 I(\text{Pclass} = 2) + \beta_3 I(\text{Pclass} = 3) + \beta_4 \text{Age}.$$



6.2 Logistic Regression Model

Example 6.2 - Solution

- Applying the `glm()` function in R, we obtain the estimated regression coefficients:

```
Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    3.65544    0.36264  10.080 < 2e-16 ***
Sexmale       -2.58426    0.18717 -13.807 < 2e-16 ***
as.factor(Pclass)2 -1.18823    0.26020  -4.567 4.96e-06 ***
as.factor(Pclass)3 -2.44234    0.24861  -9.824 < 2e-16 ***
Age           -0.03543    0.00706  -5.019 5.20e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- A natural question is how to interpret these estimated regression coefficients?
 - Interpretations under the linear regression model is straightforward, e.g., in Example 6.1, the fitted model is $Y = -266.5344 + 6.1376X + \varepsilon$, suggesting that, **on average**, a person who is 1 inch taller weights 6.1376 pounds more.
 - Interpretations under the logistic regression model depends on a concept: **odds** (赔率).



6.2 Logistic Regression Model

- Parameter interpretation under the logistic regression model:
 - Odds is the ratio of the probability of an event occurring to that of it not occurring, i.e., $p/(1 - p)$, which can be understood as the relative likelihood of the event happening.
 - If the odds of an event is greater than 1, it means the event is more likely to happen than not.
 - The concept of odds dates back to gambling and betting activities, particularly in betting on horse raising and football.
 - In these contexts, odds represent the payout ratio.
 - Later, this concept was adopted in Statistics, especially with the development and widespread use of logistic regression.
 - In logistic regression, the estimated $\hat{\beta}_j$ indicates the change in log-odds of the event happening for a one-unit change in X_j .
 - If $\hat{\beta}_j > 0$, it means the log-odds of the event happening is $\hat{\beta}_j$ higher, or equivalently, the odds of the event happening is multiplied by $\exp(\hat{\beta}_j)$.



6.2 Logistic Regression Model

Example 6.2 - Solution

- Based on the model fitting results, we have the following interpretations:
- All else being equal, the log-odds of survival for males was 2.58426 lower than that for females, that is the odds of survival for males was only $\exp(-2.58426) \approx 0.0755$ times the odds for females. Or we could say that the odds of survival for females was $\exp(2.58426) \approx 13.3$ times that for males.
- All else being equal, the log-odds of survival for 2nd class passengers was 1.18823 lower than that for 1st class passengers, i.e., the odds of survival for 1st class passengers was $\exp(1.18823) \approx 3.3$ times that for 2nd class passengers.
- All else being equal, the odds of survival for 1st class passengers was $\exp(2.44234) \approx 11.5$ times that for 3rd class passengers.
- All else being equal, for every 1-year increase in age, the log-odds of survival is lower by 0.03543, i.e., for 1-year decrease in age, the odds of survival become $\exp(0.03543) \approx 1.04$ times the original odds.
- Please note that these interpretations are solely based on the current model's fitting results. If additional explanatory variables are included in the model, the estimated parameter values would differ so that the values in the interpretations would change accordingly.



6.2 Logistic Regression Model

- Finally, let's see how to evaluate a binary classification model with quantitative measures.
 - For any given input $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})$, the logistic regression model does not output $\hat{y}_i = 0$ or $\hat{y}_i = 1$ directly, but $\hat{p}_i$, which is the estimated value of $p_i = P(Y_i = 1|\mathbf{x}_i)$.
 - Therefore, a **threshold (阈值) c** ($0 < c < 1$) need to be specified so that $\hat{y}_i = 1$ when $\hat{p}_i \geq c$ and $\hat{y}_i = 0$, otherwise.
 - Then, we could compare $\hat{y}_i$ with the observed y_i , and obtain the following results, termed the **confusion matrix (混淆矩阵)**.

Observed	Model Predicted	
	$\hat{y}_i = 1$	$\hat{y}_i = 0$
$y_i = 1$	TP	FN
$y_i = 0$	FP	TN

- Based on the confusion matrix, several quantitative measures can be defined, e.g., **准确率(accuracy)**:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{n} \times 100\%.$$

- When the two categories are highly imbalanced in the data, accuracy is not a proper measure.



6.2 Logistic Regression Model

Observed	Model Predicted	
	$\hat{y}_i = 1$	$\hat{y}_i = 0$
$y_i = 1$	TP	FN
$y_i = 0$	FP	TN

- Therefore, there are other commonly used measures other than accuracy:

- Precision (精确率): proportion of Y predicted to be 1 that is truly 1.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\%.$$

- Recall (召回率, also called sensitivity, 敏感性): proportion of $Y = 1$ correctly predicted to be 1

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\%.$$

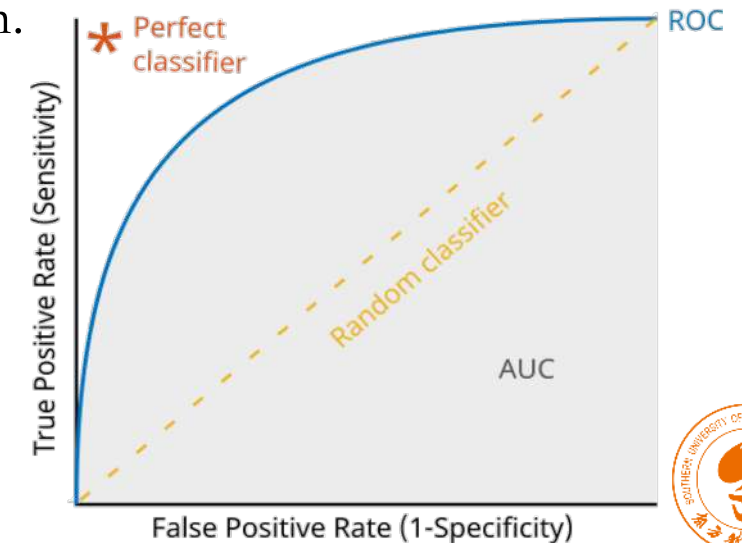
- Specificity (特异性): proportion of $Y = 0$ correctly predicted to be 0.

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \times 100\%.$$



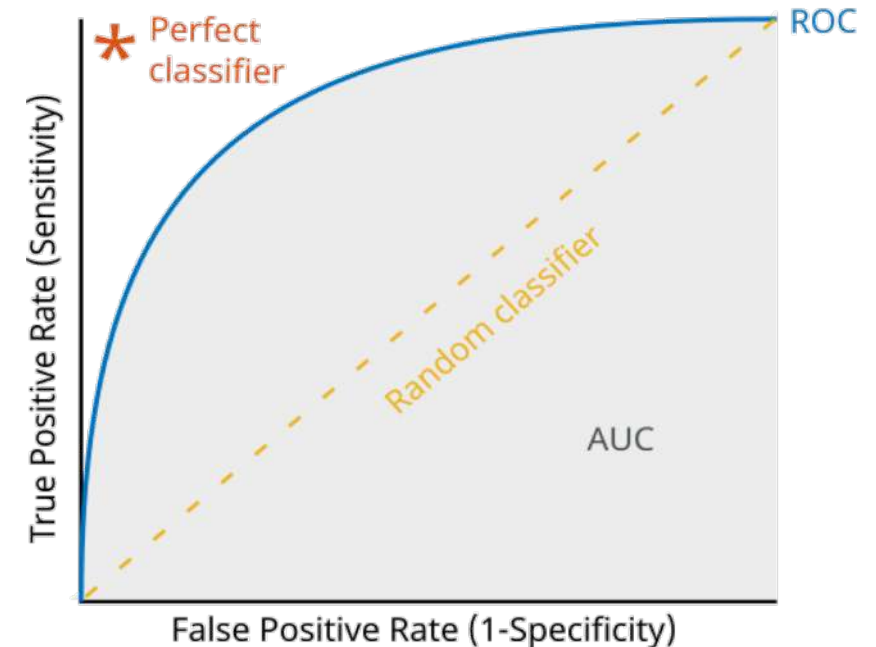
6.2 Logistic Regression Model

- The three measures are typically used in two combinations:
 - Precision and Recall: the accuracy and completeness of the model for identifying positive cases.
 - Sensitivity (Recall) and Specificity: the ability of the model to identify positive and negative cases.
- Similar to the Type I and Type II errors in hypothesis testing, when different thresholds c are used, sensitivity and specificity are traded-off against each other.
- Therefore, it is proposed to draw a curve based on the sensitivity and specificity under different thresholds, which is the famous ROC curve in binary classification.
 - ROC curve (Receiver Operating Characteristic curve) dates back to World War II, when it was first used in radar signal processing.
 - After the war, the concept was introduced to other fields, especially psychology, to evaluate the diagnostic ability of different tests.
 - In the 1960s to 1970s, ROC curve is used in medical studies to evaluate the performance of medical diagnostic tests.



6.2 Logistic Regression Model

- ROC curve:
 - With the development of machine learning, the ROC curve and its AUC (area under the curve) have become one of the standard measure for evaluating the performance of binary classification models.
 - The diagonal line in the figure represents the result of a completely random classifier.
 - The AUC of a completely random classifier is 0.5. The higher the AUC, the better the performance of the model.
 - AUC enables comparison of different models without specifying a threshold.
 - Next, let's evaluate the model based on the Titanic data in RStudio.



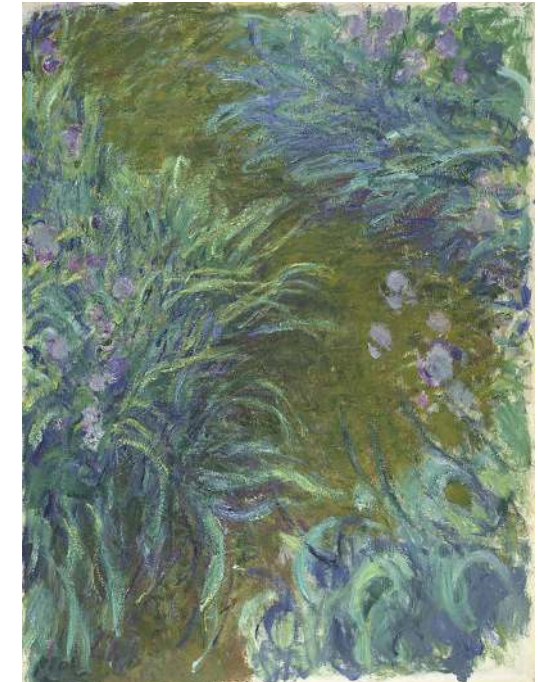
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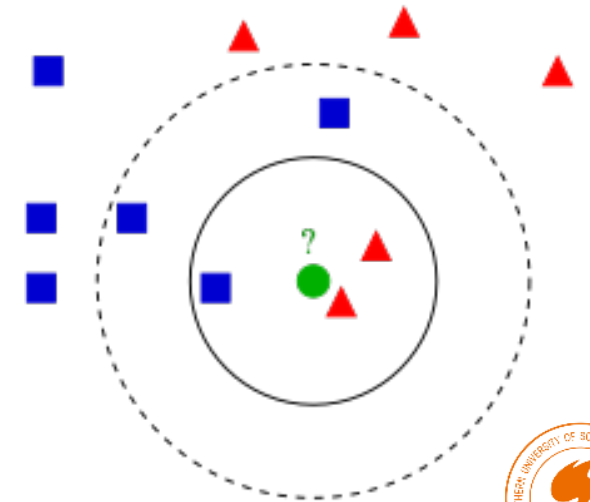
6.3 k-Nearest Neighbor Algorithm

- The iris (鸢尾花) is named after its petals, which are shaped like a kite's (鸢鸟) tail. It is loved by the French and has become the national flower of France.
- Iris was favored by many artists, e.g., Van Gogh, Hokusai, and Monet.
- The Iris dataset is an old and classic example data in the field of machine learning.



6.3 k-Nearest Neighbor Algorithm

- There are 150 records in the Iris dataset, including three types of irises, namely *Setosa* (山鸢尾), *Versicolor* (杂色鸢尾), and *Virginica* (维吉尼亚鸢尾).
- Each record contains four features, namely the length and width of the sepals and petals.
- The goal is to classify an iris based on the four features using statistical/machine learning methods.
- Here we introduce a readily comprehensible method – *k-nearest neighbor* (k-NN, k近邻) algorithm.
- The core idea of the method is the “friendship principle” – essentially capturing the notion that “birds of a feather flock together” (物以类聚, 人以群分).
 - k-nearest neighbors, as the name suggests, refers to the k nearest neighbors.
 - To determine the class of a new instance, we examine its k-nearest neighbors and assign the majority class among them.
 - While k-NN can also be applied to regression problems (taking the average of the k nearest neighbors), we focus on classification problems here.



The Iris Dataset: <https://archive.ics.uci.edu/dataset/53/iris>



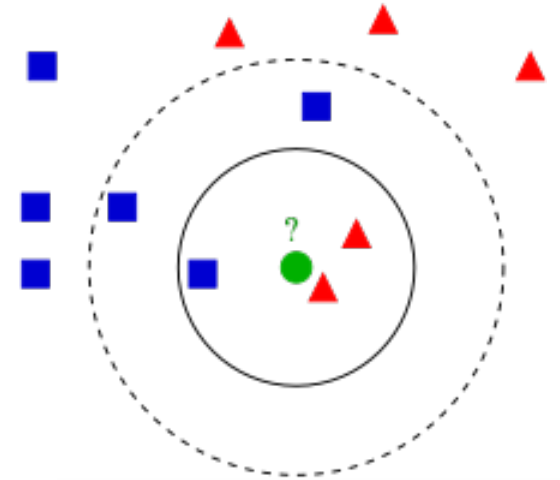
6.3 k-Nearest Neighbor Algorithm

- In practical implementation of the k-NN algorithm, several questions naturally arise:
 - What distance metric should be used to measure the similarity between instances?
 - How should we select the optimal value for k?
- The three essential components of k-NN are: **distance metric selection**, **k-value determination**, **decision rule formulation**.
- Let $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})^\top$ and $\mathbf{x}_j = (x_{j1}, x_{j2}, \dots, x_{jp})^\top$ be two feature vectors, when the features are continuous, a commonly used distance metric is the **Minkovski Distance**:

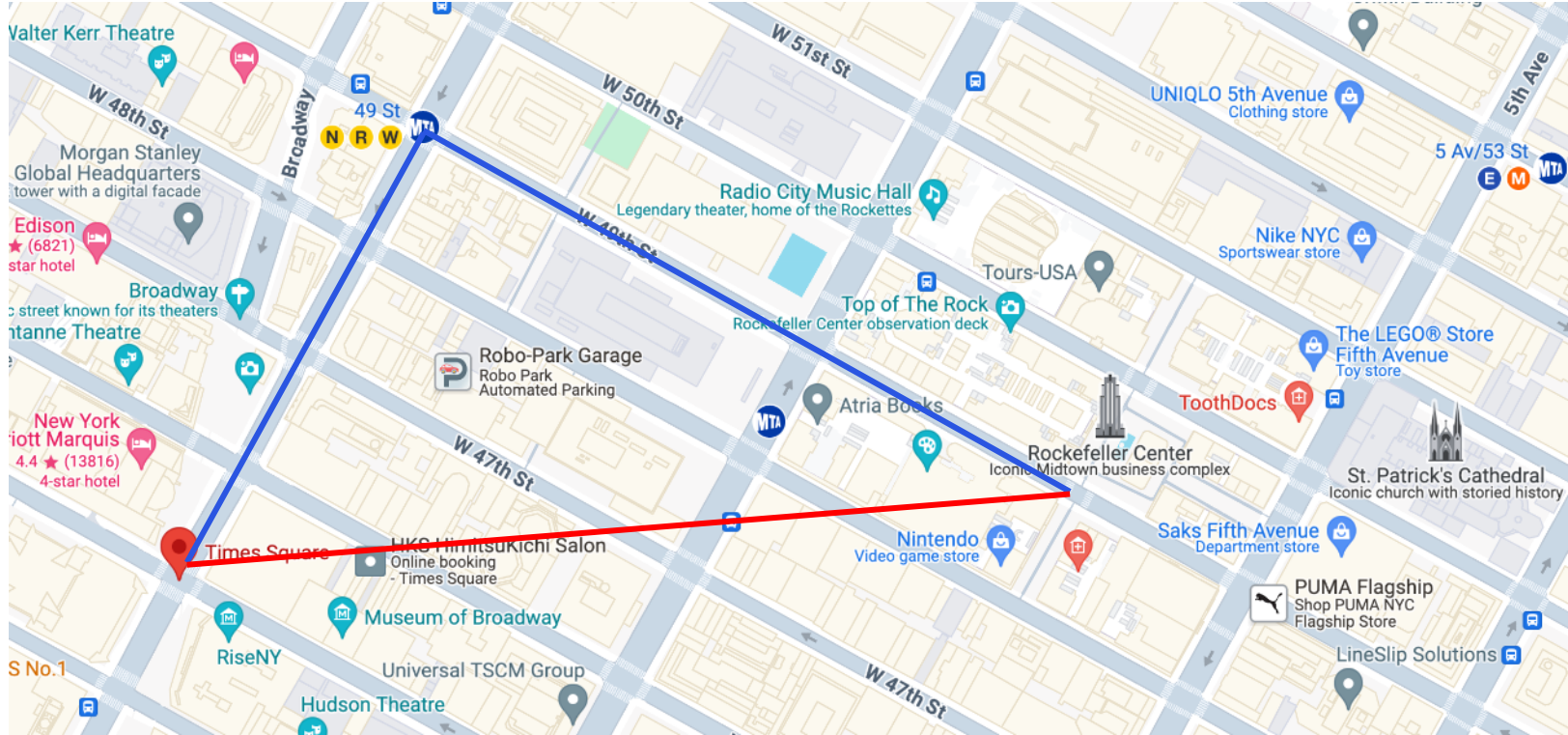
$$D(\mathbf{x}_i, \mathbf{x}_j) = \left(\sum_{l=1}^p |x_{il} - x_{jl}|^q \right)^{1/q}.$$

- When $q = 1$, it is also called the **Manhattan Distance** (曼哈顿距离):

$$D(\mathbf{x}_i, \mathbf{x}_j) = \sum_{l=1}^p |x_{il} - x_{jl}|.$$



6.3 k-Nearest Neighbor Algorithm



- When $q = 2$, it is also called the **Euclidean Distance (欧式距离)**:

$$D(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{\sum_{l=1}^p |x_{il} - x_{jl}|^2}.$$

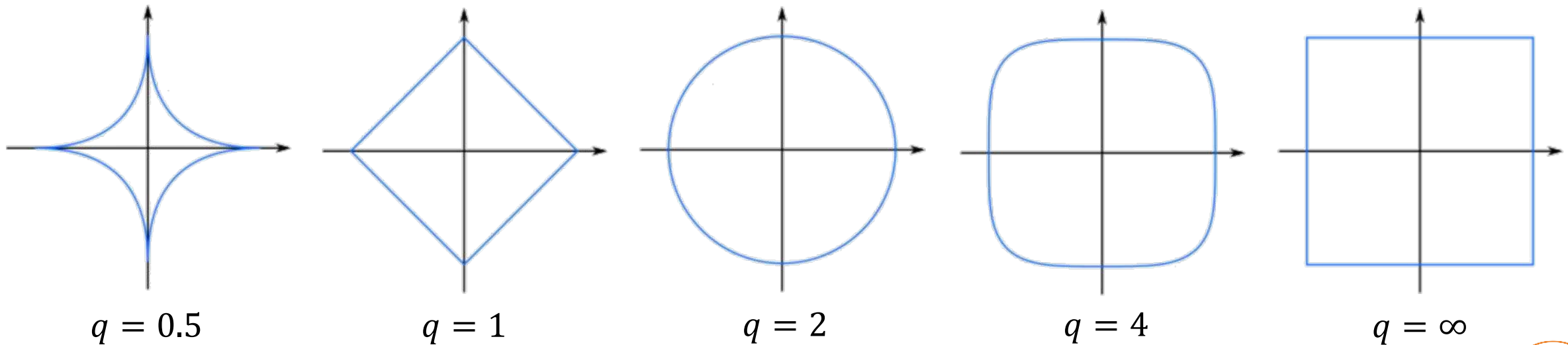


6.3 k-Nearest Neighbor Algorithm

- When $q = \infty$, it is also called the **Chebyshev Distance** (切比雪夫距离):

$$D(\mathbf{x}_i, \mathbf{x}_j) = \max_{l \in \{1, 2, \dots, p\}} |x_{il} - x_{jl}|.$$

- Taking 2D space as an example, the figure below shows the unit circle under the Minkovski distance with different values of q :



6.3 k-Nearest Neighbor Algorithm

- When using the distance metrics, it should be noted that feature scales can significantly impact the calculation results.
- To eliminate the impact of scale, each feature should be **standardized** (标准化, **normalized/scaled**).
- Commonly used standardization:

$$\tilde{x}_{il} = \frac{x_{il} - \bar{x}_l}{s_l} \quad \text{or} \quad \tilde{x}_{il} = \frac{x_{il} - \min_{j=1,\dots,n} \{x_{jl}\}}{\max_{j=1,\dots,n} \{x_{jl}\} - \min_{j=1,\dots,n} \{x_{jl}\}}.$$

- Additionally, when dealing with discrete feature - particularly in text analysis or image processing - the **Hamming distance** (汉明距离) is commonly employed.
 - The Hamming distance measure the difference between two strings of equal length, originally introduced for error detection and correction in information transmission.
 - It is defined as the number of positions where the characters of the two strings mismatch. E.g., the Hamming distance between “karolin” and “kathrin” is 3.
 - Hamming distance finds broad applications across various domains, e.g., in bioinformatics, it is used to compare DNA sequences to assess genetic similarity.



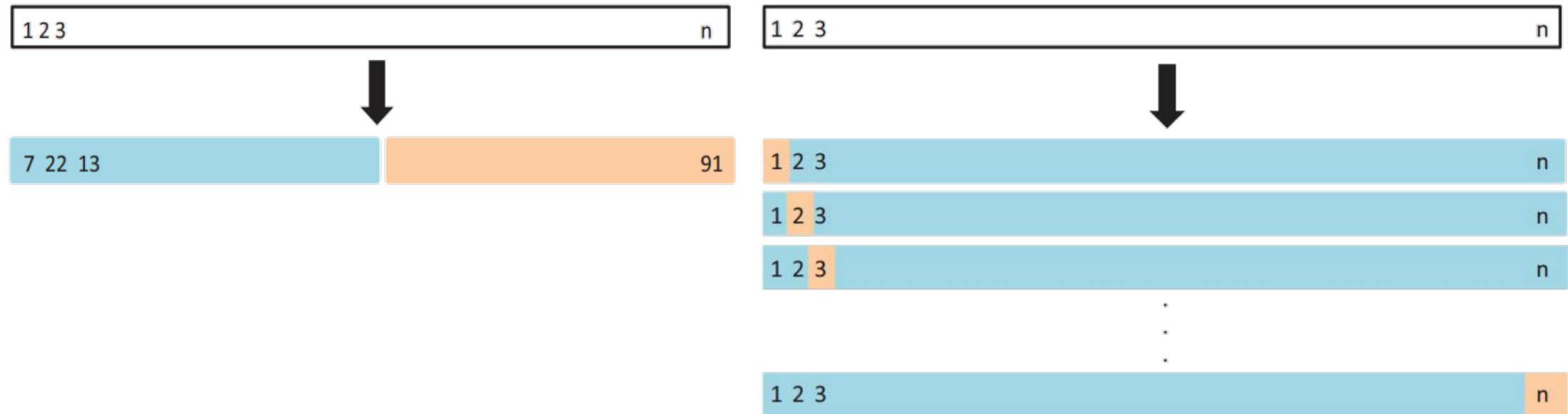
6.3 k-Nearest Neighbor Algorithm

- For decision rule, k-NN typically employs the **majority voting rule** (多数投票规则).
 - However, when dealing with imbalanced class distribution, special strategies are required to mitigate bias toward the majority class, e.g., assigning higher weights to neighbors from minority classes.
- Finally, let's talk about k-value determination.
 - Advantage of small k-values: captures fine-grained patterns and local structures in the available instances (typically called the **training data**, 训练数据). **Overfitting (过拟合)**
 - Risk of small k-values: over-sensitivity to noise in the data and poor generalization to new instances (typically called the **test data**, 测试数据).
 - Advantage of large k-values: smoother decision boundaries and reduced sensitivity to noise in the data.
 - Risk of large k-values: over-simplification (e.g., $k=n$ degenerates to predicting only the majority class).
 - A commonly used method to determine k is **cross-validation** (CV, 交叉验证). **Underfitting (欠拟合)**



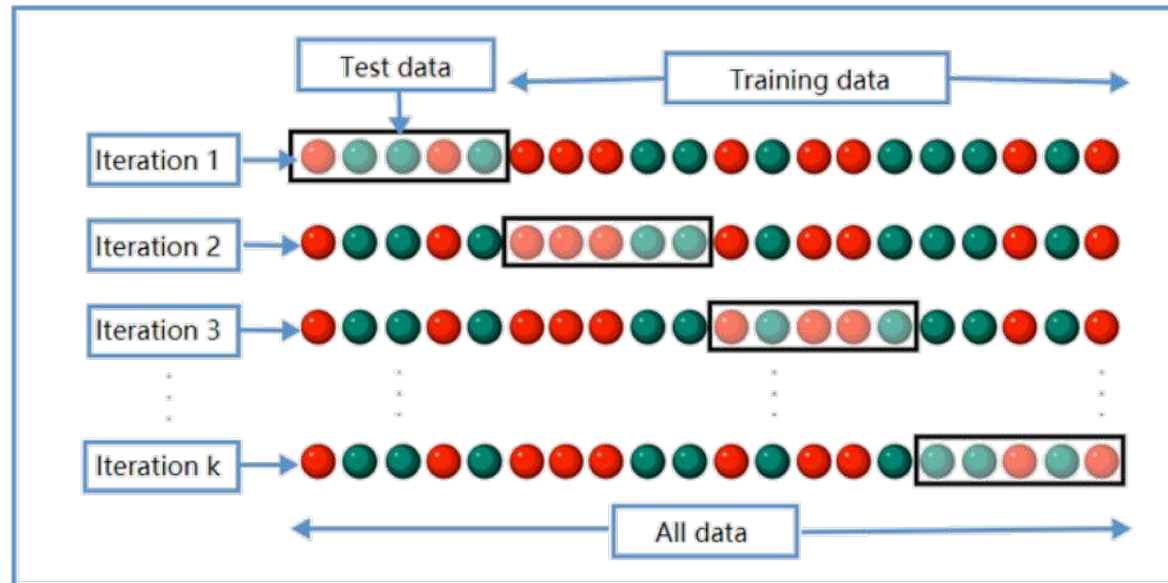
6.3 k-Nearest Neighbor Algorithm

- Cross-validation:
 - In machine learning, to evaluate the prediction performance of a model, data are typically divided into two parts, i.e., the training data and the test data.
 - However, it depends on how the data are partitioned. To reduce the impact of data partition, cross-validation is proposed.
 - There are different implementations of CV, e.g., [LOOCV](#)(leave-one-out cross validation), where each time one instance is taken as the test data and the remaining instances as the training data.



6.3 k-Nearest Neighbor Algorithm

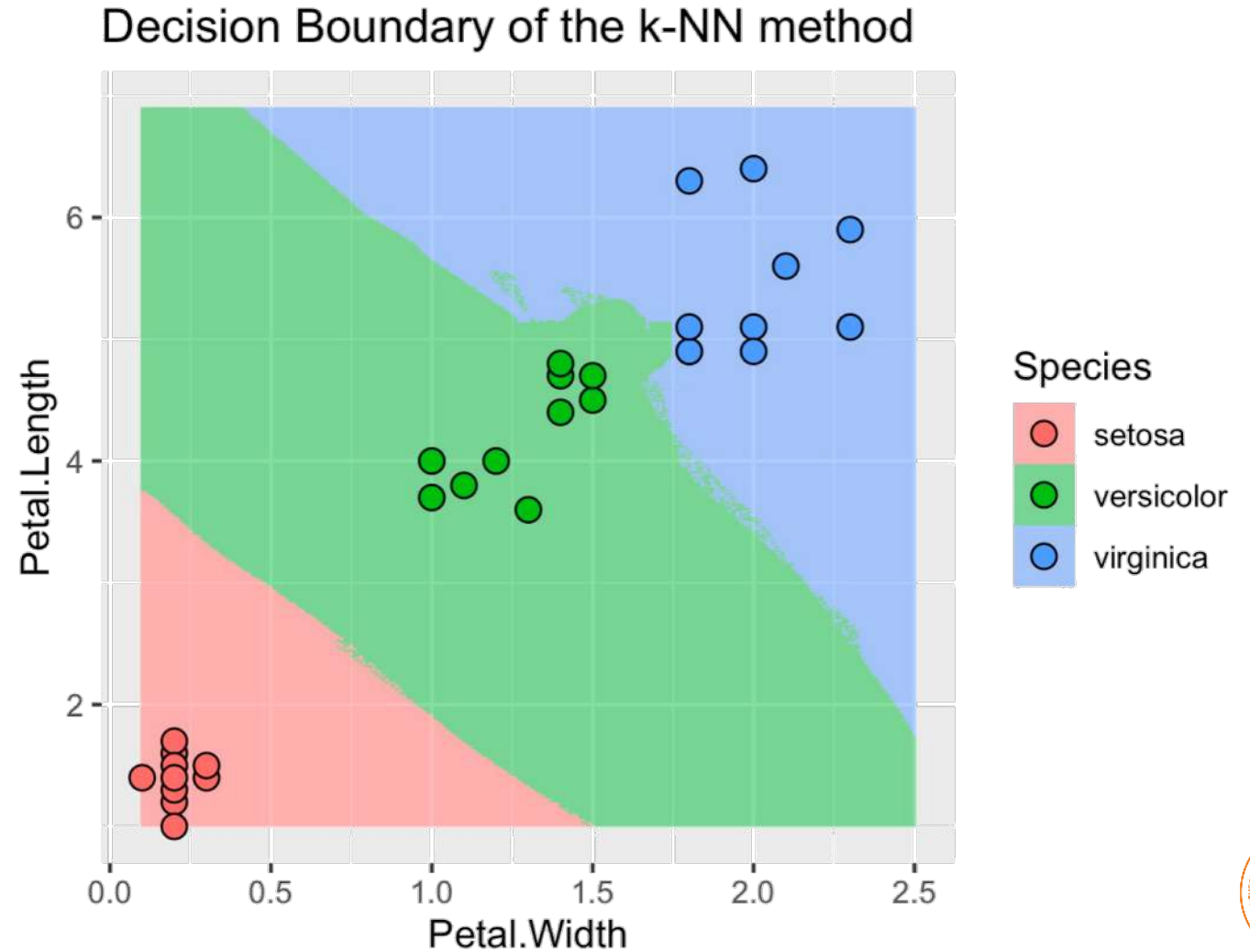
- Cross-validation:
 - LOOCV has an obvious deficiency, i.e., it is not computationally efficient. Consequently, **k-fold CV** (k折交叉验证) is proposed.
 - It splits the data into k groups. For each group, take the group as the test data and the remaining groups as the training data.
 - Combine the k evaluation results into a single evaluation score.



6.3 k-Nearest Neighbor Algorithm

Example 6.3

Implement the k-NN method on the Iris dataset in RStudio.



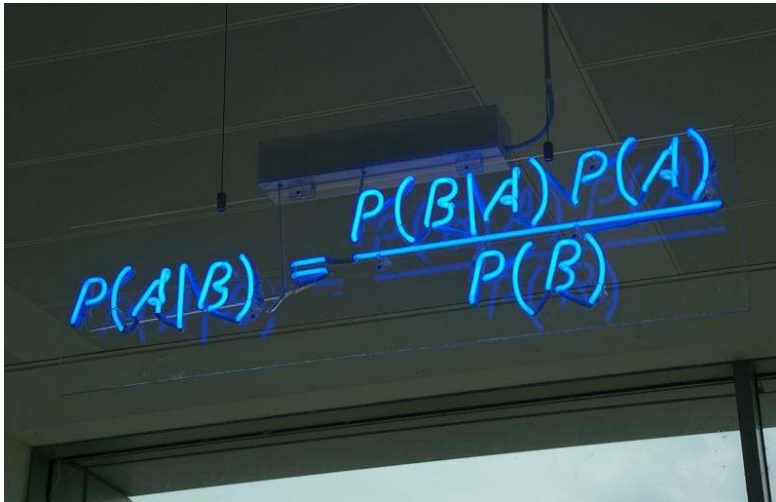
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6.4 Naïve Bayes Classifier

- Bayesian classifier (贝叶斯分类器) is a generic term for a class of classification methods, all of which are based on the **Bayes Theorem**.
- The Naïve Bayes classifier (朴素贝叶斯分类器) is the simplest and most common one.


$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$\Rightarrow P(\text{class}|\text{features}) = \frac{\text{Conditional Probability} \times \text{Prior Probability}}{P(\text{features})}$$

Conditional Probability

Prior Probability

Posterior Probability

- To obtain the posterior prob., we only need to calculate the prior prob. and conditional prob.
- In the following, we take the Titanic data as an example to explain the Naïve Bayes Classifier.
- Again, only consider three features, i.e., sex, passenger class, and age.



6.4 Naïve Bayes Classifier

- Part of the data are shown on the right. How to predict the survival of a 2nd class young male?
- What we want to obtain is $P(\text{Survived} = 1 | \text{Sex} = \text{male}, \text{Pclass} = 2, \text{AgeCat} = \text{youth})$.
- First, consider the prior probability:
 $P(\text{Survived} = 1) = 0.384, P(\text{Survived} = 0) = 0.616$.
- Then consider the conditional probability:
 $P(\text{Sex} = \text{male}, \text{Pclass} = 2, \text{AgeCat} = \text{youth} | \text{Survived} = 1)$,
 $P(\text{Sex} = \text{male}, \text{Pclass} = 2, \text{AgeCat} = \text{youth} | \text{Survived} = 0)$.
- The Naïve Bayes classifier assumes **conditional independence** (条件独立性):
 $P(\text{Sex} = \text{male}, \text{Pclass} = 2, \text{AgeCat} = \text{youth} | \text{Survived})$
 $= P(\text{Sex} = \text{male} | \text{Survived}) \times P(\text{Pclass} = 2 | \text{Survived}) \times P(\text{AgeCat} = \text{youth} | \text{Survived})$.
- This is a strong assumption which assumes that the features are independent given the class, this is the reason why the classifier is called “naïve”.

	Survived	Sex	Pclass	AgeCat
1	0	male	3	youth
2	1	female	1	middle-aged
3	1	female	3	middle-aged
4	1	female	1	middle-aged
5	0	male	3	middle-aged
6	0	male	3	middle-aged
7	0	male	1	senior
8	0	male	3	child
9	1	female	3	middle-aged
10	1	female	2	teenager



6.4 Naïve Bayes Classifier

- It can be obtained from the data:

$$P(\text{Sex} = \text{male} | \text{Survived} = 1) = 0.319, \quad P(\text{Sex} = \text{male} | \text{Survived} = 0) = 0.852.$$

$$P(\text{Pclass} = 2 | \text{Survived} = 1) = 0.254, \quad P(\text{Pclass} = 2 | \text{Survived} = 0) = 0.177.$$

$$P(\text{AgeCat} = \text{youth} | \text{Survived} = 1) = 0.193, \quad P(\text{AgeCat} = \text{youth} | \text{Survived} = 0) = 0.257.$$

- Therefore:

$$P(\text{Sex} = \text{male}, \text{Pclass} = 2, \text{AgeCat} = \text{youth} | \text{Survived} = 1) = 0.319 \times 0.254 \times 0.193 \approx 0.016.$$

$$P(\text{Sex} = \text{male}, \text{Pclass} = 2, \text{AgeCat} = \text{youth} | \text{Survived} = 0) = 0.852 \times 0.177 \times 0.257 \approx 0.039.$$

$$\begin{aligned} \Rightarrow P(\text{Survived} = 1 | \text{Sex} = \text{male}, \text{Pclass} = 2, \text{AgeCat} = \text{youth}) &= \frac{0.016 \times 0.384}{0.016 \times 0.384 + 0.039 \times 0.616} \\ &\approx 0.204. \end{aligned}$$

- And $P(\text{Survived} = 0 | \text{Sex} = \text{male}, \text{Pclass} = 2, \text{AgeCat} = \text{youth}) \approx 0.796$, so a 2nd class young male is classified as a non-survivor.



6.4 Naïve Bayes Classifier

- In the previous example, features are all categorical, what if we have continuous features?
- Simply put, we could assume distributions for the continuous features, then replace probability with density when calculating the conditional probability. In this case, **conditional likelihood** (条件似然) is calculated.
- The pros and cons of the Naïve Bayes classifier:
 - Pros: logically simple, easy to implement, computationally efficient, results are well interpretable.
 - Cons: conditional independence is a strong assumption that is typically not realistic in practice, leading to poor classification performance when features are strongly correlated.
- The Naïve Bayes classifier is often used for classification of text data, e.g., spam filtering, sentiment analysis, news classification, etc., it can usually be assumed that words appear independently.

Example 6.4

Based on the reviews on Yelp, apply the Naïve Bayes method to classify each review as positive or negative.



6.4 朴素贝叶斯法

Wow... Loved this place.	1
Crust is not good.	0
Not tasty and the texture was just nasty.	0
Stopped by during the late May bank holiday off Rick Steve recommendation and loved it.	1
The selection on the menu was great and so were the prices.	1
Now I am getting angry and I want my damn pho.	0
Honeslty it didn't taste THAT fresh.)	0
The potatoes were like rubber and you could tell they had been made up ahead of time being kept under a warmer.	0



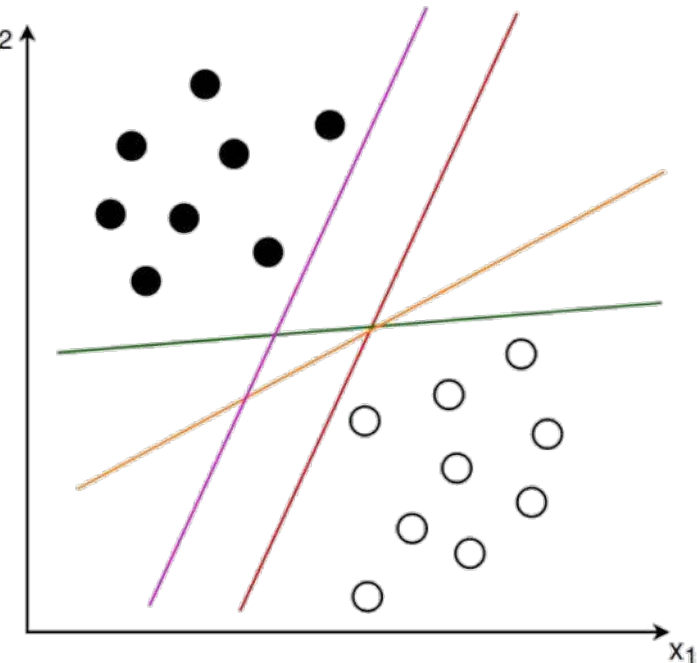
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6.5 Support Vector Machine

- Support vector machine (SVM, 支持向量机):
 - SVM once dominated the field of machine learning, maintaining its influential status for many years.
 - Even today, when neural networks being widely popular, SVM remains highly competitive on small and medium-sized datasets.
 - In simple terms, SVM is a binary classification method. Its core idea is to represent instances as data points in space and separate the two classes with the widest possible margin.
 - E.g., in the figure on the right, the goal is to find a straight line x_2 that separates the black dots from the white dots.
 - A straight line can be represented by $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b = w_1x_1 + w_2x_2 + b = 0$.
 - When $\mathbf{x}$ is a 3D vector, $f(\mathbf{x}) = 0$ is a plain in the 3D space.
 - When $\mathbf{x}$ is a $p(p > 3)$ -dim vector, $f(\mathbf{x}) = 0$ is a $(p - 1)$ -dim **hyperplane** (超平面) in the p -dim space.
 - Data that can be perfectly divided into their respective classes by a hyperplane is called **linearly separable** (线性可分的).



6.5 Support Vector Machine

- Such a hyperplane may not be unique, so how do we determine the optimal one?
- A natural intuition is that the best hyperplane should have the **maximum margin** (最大间隔).
- Instances from both classes lie on opposite sides of the hyperplane, and the distance from the closest instances to the hyperplane is maximized.
- The data points closest to this hyperplane are called **support vectors** (支持向量).

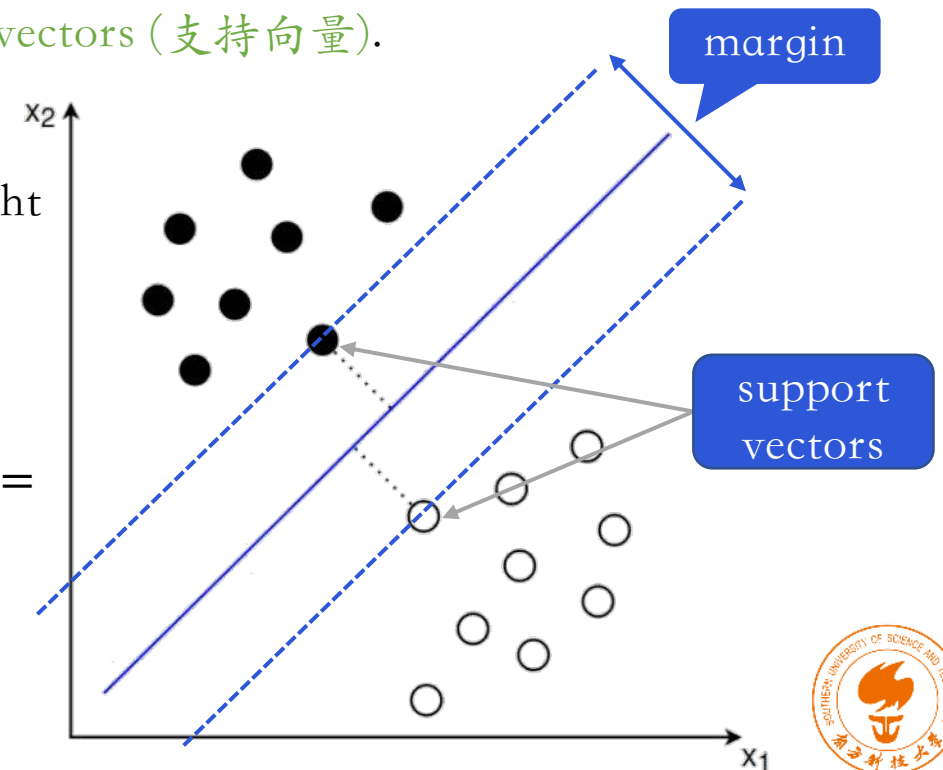
- Next, we formalize this idea mathematically.

- In a 2D space, the distance from a point (x_{01}, x_{02}) to a straight line $w_1x_1 + w_2x_2 + b = 0$ is

$$\frac{|w_1x_{01} + w_2x_{02} + b|}{\sqrt{w_1^2 + w_2^2}}.$$

- Generalized to the p -dim space, the distance from a point $\mathbf{x}_0 = (x_{01}, x_{02}, \dots, x_{0p})$ to a hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$ is

$$\frac{|\mathbf{w}^T \mathbf{x}_0 + b|}{\sqrt{w_1^2 + w_2^2 + \dots + w_p^2}} = \frac{|\mathbf{w}^T \mathbf{x}_0 + b|}{\|\mathbf{w}\|}.$$



6.5 Support Vector Machine

- The mathematical expression of maximum margin:
 - For the training data $\{(y_i, \mathbf{x}_i), i = 1, 2, \dots, n\}$, let $y_i = 1$ when $\mathbf{w}^\top \mathbf{x}_i + b > 0$ and $y_i = -1$ when $\mathbf{w}^\top \mathbf{x}_i + b < 0$.
 - Looking for the hyperplane with the maximum margin is to solve the following optimization problem:

$$\max_{\mathbf{w}, b} \min_{i=1, \dots, n} 2 \times \frac{y_i(\mathbf{w}^\top \mathbf{x}_i + b)}{\|\mathbf{w}\|}.$$

- Denote $\gamma = \min_{i=1, \dots, n} y_i(\mathbf{w}^\top \mathbf{x}_i + b)$, then it is equivalent to a constrained optimization problem:

$$\begin{aligned} \max_{\mathbf{w}, b} \quad & \frac{2\gamma}{\|\mathbf{w}\|} \\ \text{s. t.} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq \gamma, i = 1, \dots, n. \end{aligned}$$

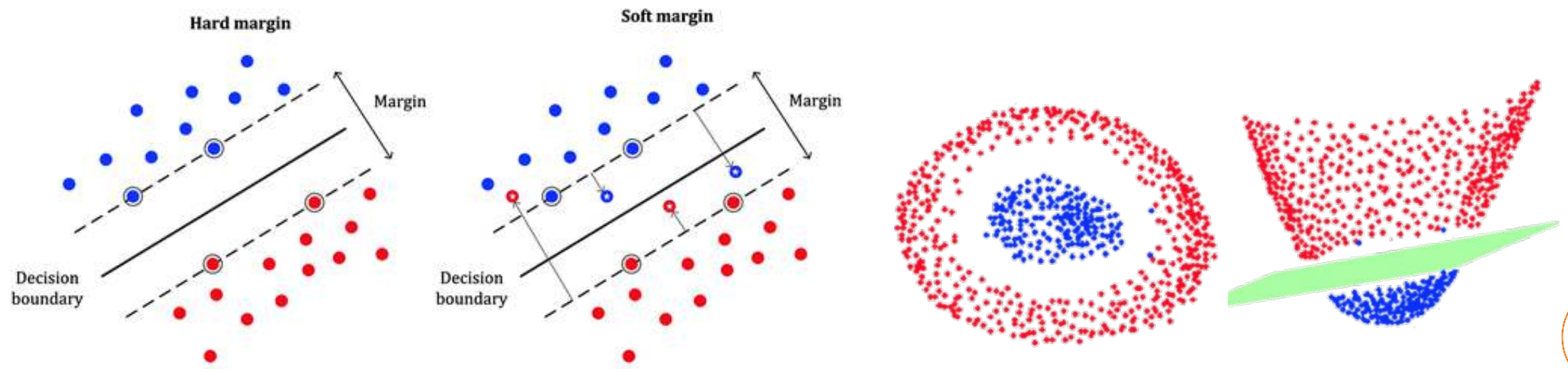
- As the value of γ does not impact the solution of the problem, without loss of generality, let $\gamma = 1$ (understood as the scaling of the coordinate system), the optimization problem of SVM is expressed as:

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s. t.} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, i = 1, \dots, n. \end{aligned}$$



6.5 Support Vector Machine

- The solution of the optimization problem is relatively complex and not further discussed here.
- SVM has many generalizations.
 - In practical applications, linearly separable data is rare.
 - When dealing with **approximately linearly separable** data, researchers proposed the **soft margin** (软间隔) - allowing some data points to appear within the margin zone.
 - For **linearly inseparable data**, data points can be mapped into a higher-dimensional space and find the hyperplane with maximum margin, resulting in a **nonlinear SVM**.



Chapter 6 Common Supervised Learning Methods I

- 6.1 Linear Regression Model
- 6.2 Logistic Regression Model
- 6.3 k-Nearest Neighbor Algorithm
- 6.4 Naïve Bayes Classifier
- 6.5 Support Vector Machine
- 6.6 General Pattern of Supervised Learning



6.6 General Pattern of Supervised Learning

- Supervised learning (监督学习) refers to learning the correspondence between input and output from labeled data (标注数据).
- Each input is an instance (实例), typically represented by a feature vector (特征向量).
- Let $\mathbf{X}$ be the input variables and Y be the output variable, their observed values are denoted by $\mathbf{x}, y$.
- Training data (训练数据) and test data (测试数据) both consist of pairs of input feature vector and output, e.g., $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2) \dots, (\mathbf{x}_n, y_n)\}$.
- Supervised learning assumes that $\mathbf{X}$ and Y follow a joint probability distribution (联合概率分布) $P(\mathbf{X}, Y)$, and the training and test data are considered to be a simple random sample from the distribution.
- Supervised learning models can be either probabilistic or non-probabilistic:
 - Probabilistic model: represented by the conditional distribution function (条件概率分布) $P(Y|\mathbf{X})$, e.g., the linear regression model, logistic regression model, naïve Bayes algorithm, etc.
 - Non-probabilistic model: represented by the decision function (决策函数) $Y = f(\mathbf{X})$, e.g., k-NN, SVM.



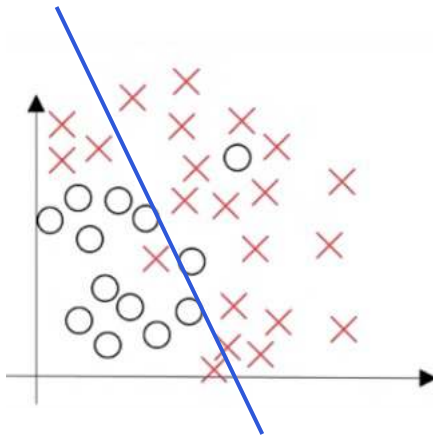
6.6 General Pattern of Supervised Learning

- Supervised learning fit/learn a model (estimate the parameters in the model) to the training data, and apply the fitted model to make predictions on the test data.
- Statistical learning/machine learning methods are composed of three core components, which can be summarized as: **Method = Model + Strategy + Algorithm** (方法=模型+策略+算法).
 - **Model** is essentially the **conditional distribution function** or the **decision function** to be learned.
 - Once a model is defined, the next step is to determine criteria for learning or selecting the optimal model.
 - This is typically achieved by minimizing a **loss function** (损失函数) or **cost function** (代价函数).
 - **Strategy** refers to how the learning problem is transformed to an optimization problem.
 - However, optimization problems may lack explicit analytical solutions, requiring the application of numerical computation methods.
 - **Algorithm** corresponds to the concrete computational procedures employed during the learning process.

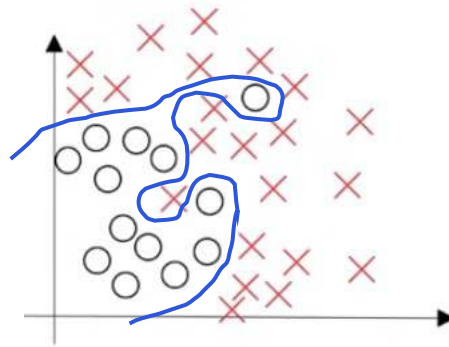


6.6 General Pattern of Supervised Learning

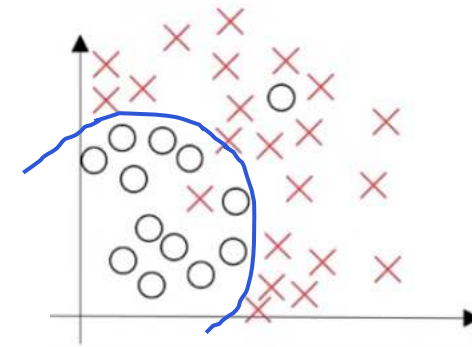
- Lastly, let's discuss the **bias-variance tradeoff** phenomenon in supervised learning.
- Typically, as the model complexity increases (e.g., the number of parameters increases), the **bias** (偏差) and **variance** (方差) of the model show a trade-off relationship.



High bias
Underfitting
(欠拟合)



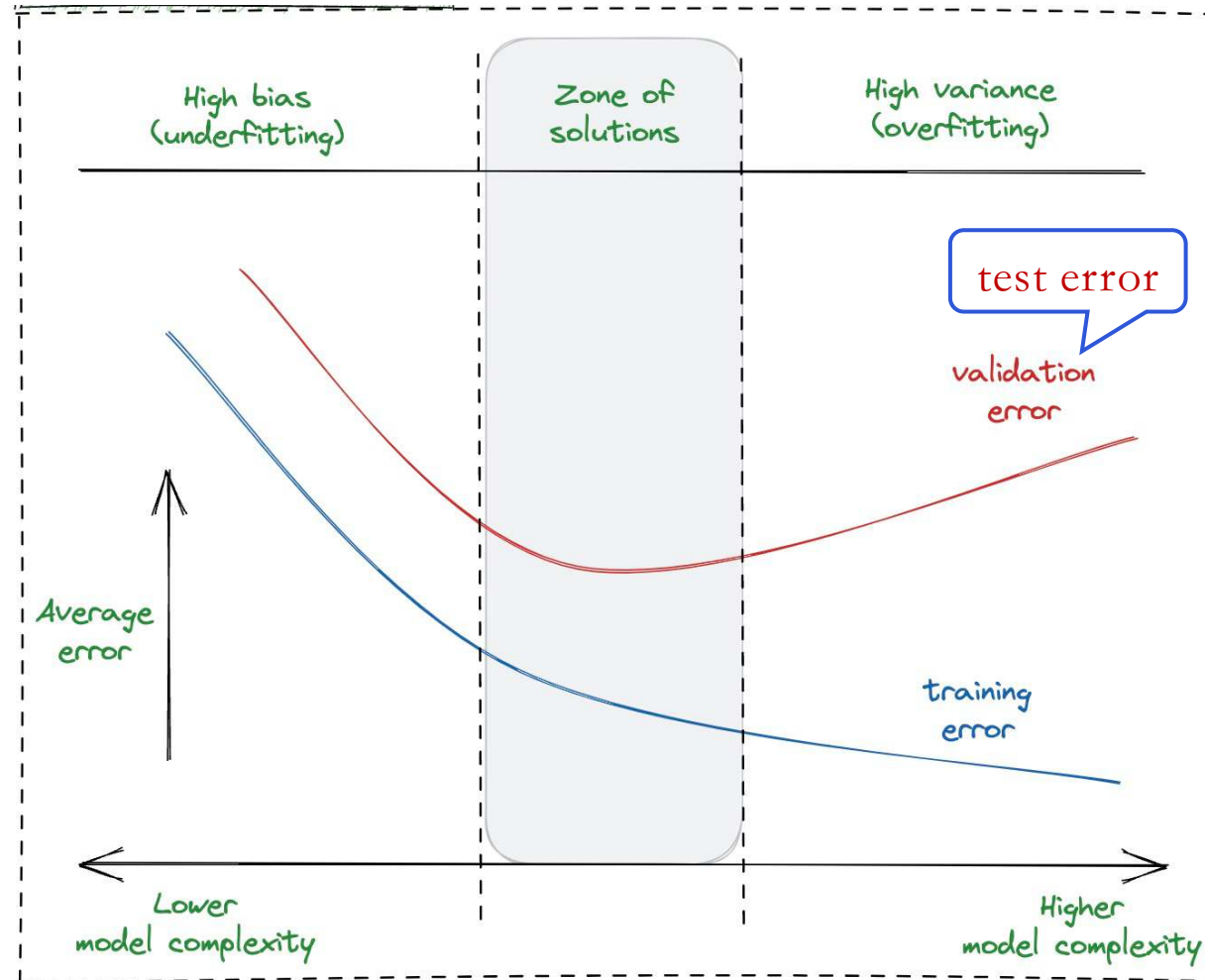
High variance
Overfitting
(过拟合)



“Just right”

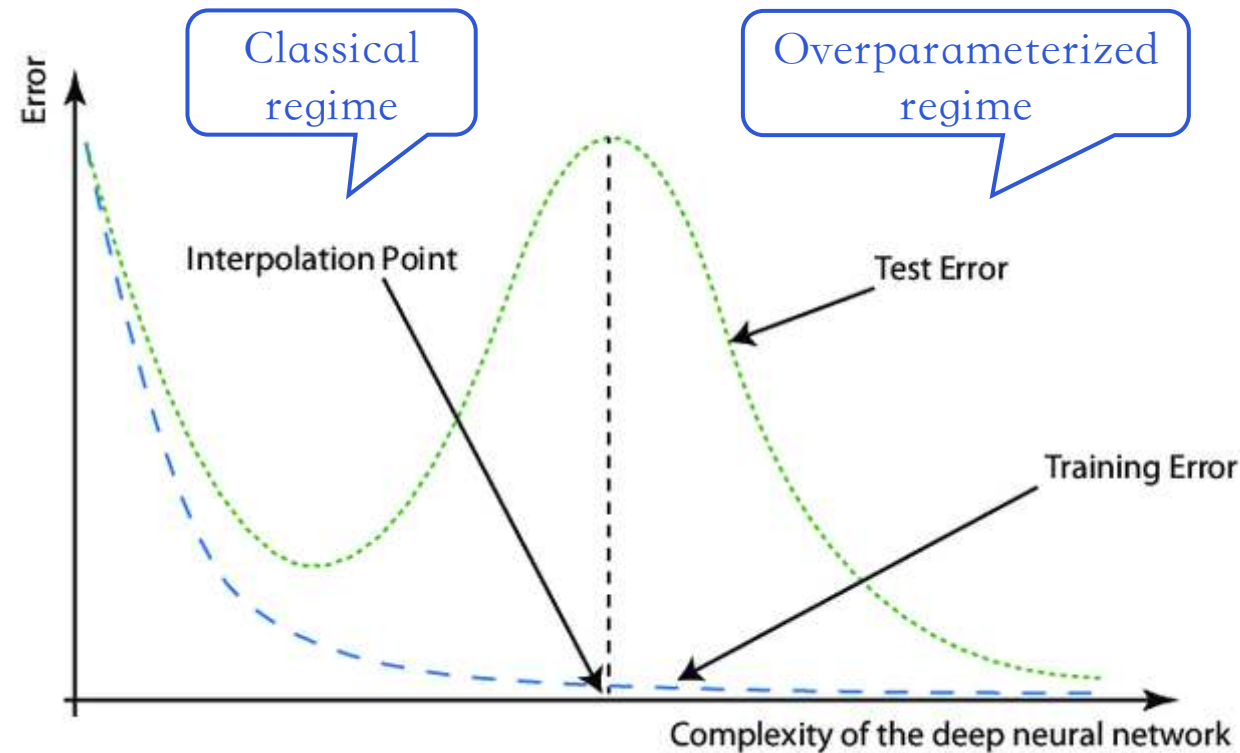
6.6 General Pattern of Supervised Learning

- How to choose a model with appropriate complexity in practice?



6.6 General Pattern of Supervised Learning

- In recent years, researchers observed a phenomenon that is not completely consistent with the traditional bias-variance tradeoff, which is called the **double descent phenomenon**.





End of Chapter 6